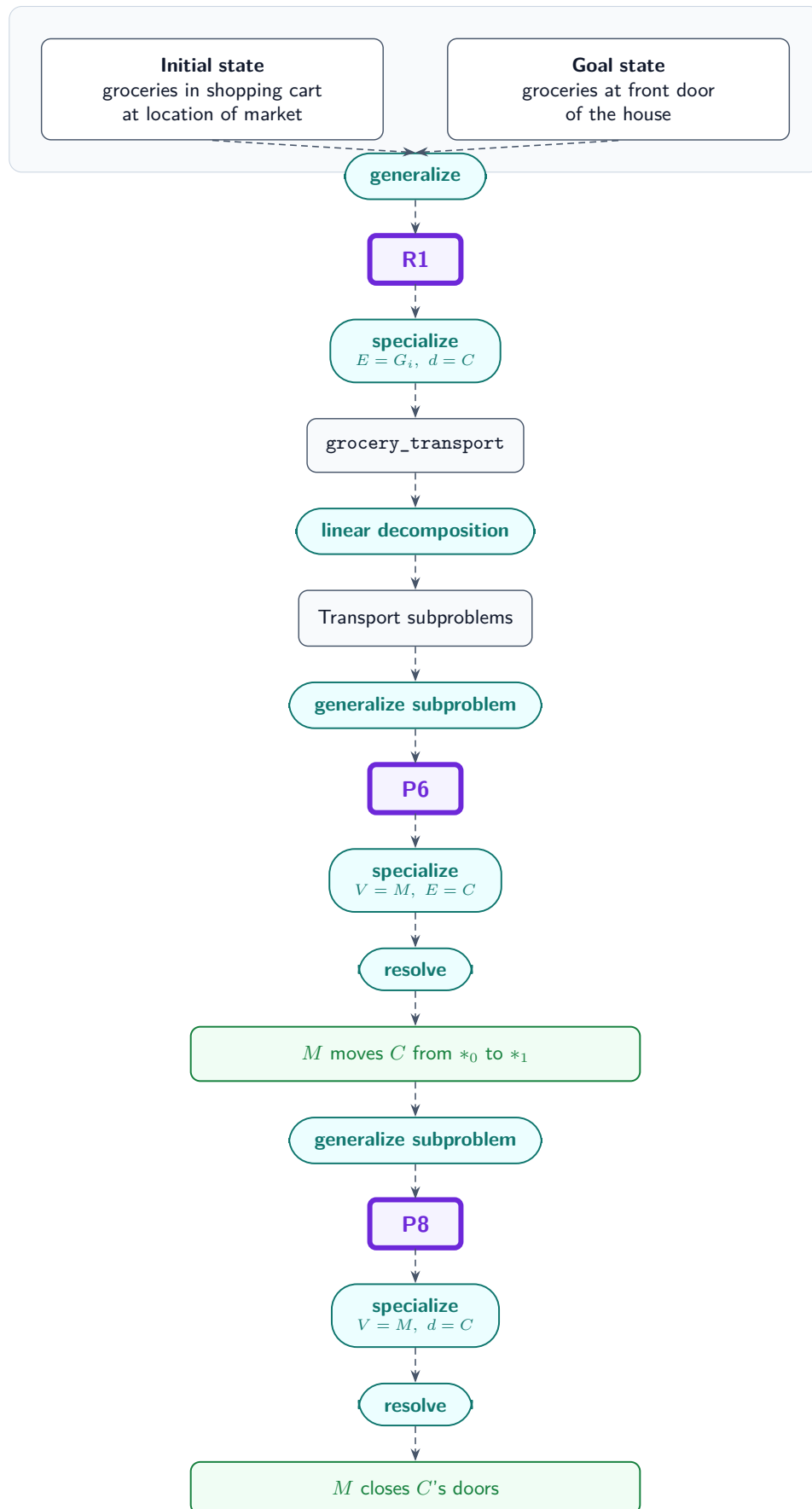


Grocery-delivery scenario



R1 · relocate an object

Parameters E = object, d = carrier

Problem Initial: E at $*_1$ Goal: E at $*_2$

linear decomposition

1. E at $*_1$
2. d at $*_1$
3. E in d
4. E at $*_2$

P6 · move an object with a carrier

Parameters E = object, V = carrier

Problem Initial: $L_E = *_1$ Goal: $L_E = *_2$

linear decomposition

1. $L_E = *_1$
2. $L_E = *_1$; $L_V = L_E$
3. $L_E = *_1$; $L_V = L_E$; V connected to E
4. $L_E = *_1$; $L_V = L_E$; V connected to E ; safety check
5. $L_E = *_2$

resolution

Source: state 4

Action: V moves E from $*_1$ to $*_2$

Target: state 5

P8 · close a vehicle's doors

Parameters V = mover, d = vehicle

Problem Initial: d 's doors open Goal: d 's doors closed

linear decomposition

1. d 's doors open
2. d 's doors open; d within reach of V
3. d 's doors closed

resolution

Source: state 2

Action: V closes d 's doors

Target: state 3